

MAINTENANCE

PART 2 — PLAN

CASE

FILE

Activity 05 — Maintaining Your Own System · Day 2

Expediente de mantenimiento — Parte 2: plan y control de cambios

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System / Sistema	Entre Estrellas - Cafetería y Souvenirs
Version before / Versión actual	v1.0
Version after / Versión resultante	v1.1
Maintenance window / Ventana	Fri 4 Sep, 08:00 – 09:30
Date / Fecha	2026-09-11
Teacher / Docente	Ing. Javier Arturo García Mineros

1. Change requests

1. Solicitudes de cambio

ID	FINDING	WHAT IS REQUESTED (ENGLISH)	REQUESTED BY	TYPE	DECISION AND REASON
CR-01	F-01	Restrict the Cashier role so it can no longer create, edit or permanently delete products (name, price, stock). The only action a Cashier can perform on a product is toggling it between Active and Inactive. Full product management (create, edit, delete) becomes exclusive to the Administrator role.	Product owner / development team	Corrective	Approved. Highest-severity open finding (High); low implementation effort; closes a real security/accountability gap before more staff use the system.

CR-02	F-03	Translate the entire user interface (login, dashboard, POS, inventory, suppliers, employees, reports, kiosk) from Spanish to English, keeping the original term only where there is no direct English equivalent (e.g., DUI).	Teacher / course requirement (Module 3.6)	Adaptive	Approved. Required by the current project phase; medium effort due to the number of screens, but touches no business logic, so risk stays low.
CR-03	F-06	Add an automatic expiration to kiosk orders: any order still "Pending" after 20 minutes is automatically marked "Cancelled" so it stops cluttering the cashier's queue.	Cashier role (operational need)	Corrective	Approved. Closes a real functional gap; low effort (one comparison added to an existing query).
CR-04	F-05	When a product's image is replaced, delete the previous image file from public/uploads/productos/ instead of leaving it orphaned on disk.	Development team (housekeeping)	Preventive	Approved, low priority. Small and safe; only executed if time remains after CR-01 to CR-03.
CR-05	F-04	Configure an automated nightly backup job for EntreEstrellasDB (SQL Server Agent), keeping 7 daily copies.	Development team (risk reduction)	Preventive	Deferred. Valuable but out of scope for a 3×45-min window; proposed for the maintenance cycle starting Mon 14 Sep.

2. Impact analysis

2. Análisis de impacto

CR	FILES, SCREENS AND FUNCTIONS AFFECTED	DATABASE OBJECTS AFFECTED	WHO IS AFFECTED WHILE IT IS APPLIED
CR-01	modulos/inventario/guardar_producto.php, modulos/inventario/eliminar_producto.php, modulos/inventario/inventario.php (the	None — no schema change, only the code path that reaches inventario_productos	Cashier users only; Administrator and Warehouse workflows are unchanged

	requerirRol() calls and the \$puedeEditar flag that shows/hides the edit form and buttons)		
CR-02	Every .php file under public/ and modulos/ that has visible text (labels, buttons, table headers, alert()/confirm() messages), plus the shared includes/sidebar.php	None	All three roles — every screen's language changes, so testing must cover Administrator, Cashier and Warehouse
CR-03	modulos/ventas/ventas_pos.php (pending-orders panel and its query)	pedidos_kiosko — its estado column is updated by a new auto-cancel check	Customers using the kiosk (if their order times out) and the cashier viewing the "Pedidos en espera" queue
CR-04	modulos/inventario/guardar_producto.php	None (filesystem only; imagen_url column unaffected)	Administrator/Warehouse when editing a product's image

3. Prioritization

3. Priorización

CR	SEVERITY	EFFORT	PRIORITY	ENTERS THIS MAINTENANCE? IF NOT, PROPOSED DATE
CR-01	High	Low	DO it first.	Yes
CR-02	Medium	Medium	Do it.	Yes
CR-03	Medium	Low	Do it.	Yes
CR-04	Low	Low	Do it at the end, if there is time.	Yes (best-effort)
CR-05	High	High	Split it — reduce risk now, plan the rest.	No — deferred to Mon 14 Sep

4. Procedure for each approved change

4. Procedimiento de cada cambio aprobado

CR	OBJECTIVE (MEASURABLE)	PROCEDURE — NUMBERED STEPS	TOOLS · TIME	RISK	ROLLBACK STEPS
CR-01	After the change, a Cashier session calling guardar_producto.php or eliminar_producto.php directly by URL is redirected to panel_control.php with no database write; only Administrator can save, edit or delete a product.	1. In <code>modulos/inventario/guardar_producto.php</code>, confirm <code>requerirRol()</code> allows only ['Administrador'] (remove 'Bodega' if the new rule is admin-only, per this request). 2. Do the same in <code>modulos/inventario/eliminar_producto.php</code>. 3. In <code>modulos/inventario/inventario.php</code>, set <code>\$puedeEditar = in_array(\$rolUsuario, ['Administrador'], true)</code>; so the "+ New Product" button, the edit form and the Activate/Deactivate button behave per role — but keep a separate, smaller toggle visible to Cashier that only calls a toggle-status action, not the full edit form. 4. Re-test logging in as <code>cajero/caja123</code>: product list is visible (read-only) with only an Activate/Deactivate control; no "+ New Product" or "Edit" buttons appear. 5. Confirm Administrator (<code>admin/admin123</code>) still has full create/edit/delete access.	Text editor, XAMPP local server, the 3 test accounts. Estimated time: 25 minutes.	Very low — this only tightens an existing check. Risk of doing nothing: a Cashier can currently alter or delete the product catalog by mistake or intentionally.	Restore the previous versions of the 3 edited .php files from the source-code backup .zip (Part 1, Section 2). No database change to undo.
CR-02	Every screen (login, dashboard, POS, inventory, suppliers, employees, reports, kiosk) shows its labels, buttons, table headers and alert messages in English; no Spanish user-facing text remains except untranslatable local terms (e.g., DUI).	1. List every hard-coded Spanish string per file (labels, buttons, <code>alert()</code>/<code>confirm()</code> text, table headers). 2. Translate <code>public/index.php</code> (login) and <code>includes/sidebar.php</code> first, since they appear on every screen. 3. Translate each module file one at a time: <code>panel_control.php</code>, <code>ventas_pos.php</code>, <code>inventario.php</code>, <code>proveedores.php</code>, <code>empleados.php</code>, <code>reportes.php</code>, <code>kiosko.php</code>. 4. Update JavaScript <code>alert()</code>/<code>confirm()</code> strings inside each file's <code><script></code> block. 5. Re-test each screen as each of the 3 roles right after translating it, to catch a broken string early.	Text editor, browser, 3 test accounts. Estimated time: 2.5 hours (~15 files).	Medium — many files touched raises the chance of breaking a JS string or PHP concatenation by accident; mitigated by testing screen-by-screen instead of all at the end.	Restore the previous (Spanish) versions of the edited files from the source-code backup .zip. No database change involved.

CR-03	A kiosk order that has been "Pending" for more than 20 minutes is automatically marked "Cancelled" and no longer appears in the cashier's "Pedidos en espera" panel.	1. In <code>modulos/ventas/ventas_pos.php</code>, before the query that loads pending orders, run: <code>UPDATE dbo.pedidos_kiosko SET estado='Cancelado' WHERE estado='Pendiente' AND DATEDIFF(MINUTE, fecha_creacion, GETDATE()) > 20;</code> 2. Let the existing <code>SELECT</code> that fills <code>\$pedidosPendientes</code> run afterward, so cancelled orders no longer show. 3. Add a short note on the kiosk confirmation screen: "Orders not attended within 20 minutes are automatically cancelled." 4. Test: place a kiosk order, manually set its <code>fecha_creacion</code> 21 minutes in the past via SSMS, reload Ventas/POS, confirm it disappears and its <code>estado</code> shows <code>Cancelado</code>.	Text editor, SSMS to simulate an old timestamp. Estimated time: 30 minutes.	Low — only touches rows already <code>Pendiente</code> . Risk of doing nothing: pending orders pile up forever in the cashier's queue.	Remove/comment out the added <code>UPDATE</code> ; manually restore any order wrongly cancelled during testing with <code>UPDATE dbo.pedidos_kiosko SET estado='Pendiente' WHERE id_pedido IN (...)</code> .
CR-04	Replacing a product's image deletes the previous image file from <code>public/uploads/productos/</code> instead of leaving it on disk.	1. In <code>guardar_producto.php</code>, before saving the new uploaded file on an update, <code>SELECT</code> the product's current <code>imagen_url</code>. 2. After the <code>UPDATE</code> succeeds, if the old <code>imagen_url</code> is not null and differs from the new one, call <code>unlink()</code> on the old file's full path, guarded by <code>file_exists()</code>. 3. Test: edit a product's image twice in a row; confirm only the latest file remains in the uploads folder.	Text editor, file explorer. Estimated time: 20 minutes.	Low — <code>unlink()</code> only removes a file, never a database row.	Remove the <code>unlink()</code> call; no data is lost since the database row is never touched.

5. Database change plan

5. Plan de cambios en la base de datos

STRUCTURAL CHANGE	SCRIPT OR METHOD TO APPLY IT	SCRIPT OR METHOD TO UNDO IT	EFFECT ON EXISTING DATA
Add <code>modificado_por</code> and <code>fecha_modificacion</code> to <code>inventario_productos</code> , to record which Administrator last changed a product now that editing is	<code>ALTER TABLE dbo.inventario_productos</code> <code>ADD modificado_por INT NULL,</code> <code>fecha_modificacion DATETIME NULL;</code> <code>ALTER TABLE dbo.inventario_productos</code> <code>ADD CONSTRAINT</code> <code>FK_producto_modificador FOREIGN</code> <code>KEY (modificado_por) REFERENCES</code> <code>dbo.empleados_nomina(id_empleado);</code>	<code>ALTER TABLE</code> <code>dbo.inventario_productos</code> <code>DROP CONSTRAINT</code> <code>FK_producto_modificador;</code> <code>ALTER TABLE</code> <code>dbo.inventario_productos</code> <code>DROP COLUMN</code>	Both columns accept <code>NULL</code> , so every existing product row keeps working with <code>modificado_por = NULL</code> until the next time

admin-only (supports CR-01).		modificado_por, fecha_modificacion;	an Administrator edits it — nothing breaks, no backfill required.
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Reminder: every structural change is applied to the development copy first, and the database is exported immediately before applying it to the copy in use.

Todo cambio estructural se aplica primero en la copia de desarrollo, y la base se exporta justo antes de aplicarlo en la copia en uso.

6. Test plan

6. Plan de pruebas

ID	TYPE	WHAT IS TESTED	STEPS	EXPECTED RESULT
TC-01	Change	Cashier cannot edit/delete a product	Log in as cajero/caja123, try to reach guardar_producto.php / eliminar_producto.php directly by URL	Redirected to panel_control.php; no Edit/New Product button visible; DB row unchanged
TC-02	Change	Administrator can still fully manage products	Log in as admin/admin123; create, edit and deactivate a product	All three actions succeed and reflect immediately in the list
TC-03	Change	UI text is in English	Open every screen as each of the 3 roles and read every label/button/alert	No Spanish text remains except untranslatable local terms
TC-04	Regression	Login still works	Sign in with all three test accounts	Each reaches its own panel_control.php with the correct role menu
TC-05	Regression	Sales flow (POS) still works	As cajero, add 2 products to the cart and confirm a sale	Sale is saved, stock decreases, ticket prints correctly

TC-06	Regression	Kiosk flow still works	Place a kiosk order, confirm it appears as Pending in Ventas/POS	Order number generated; cashier sees it in "Pedidos en espera"
TC-07	Regression	Existing data intact	Count rows in dbo.detalle_ventas before and after	Same count as the Part 1 baseline, plus only the sales made during testing
TC-08	Regression	Warehouse role still works	Log in as bodega/bodega123, open Inventory and Suppliers	Both screens load correctly, unaffected by the Cashier restriction

7. Window and communication

7. Ventana y comunicación

7.1 Notification to users

Dear team, Entre Estrellas will be unavailable on Friday 4 September between 08:00 and 09:30 AM while we apply a scheduled system update. Please complete and save any active sales or inventory changes before 08:00 AM. We will send an update as soon as the system is ready. Thank you for your patience.

Estimado equipo, el sistema Entre Estrellas no estará disponible el viernes 4 de septiembre entre las 08:00 y las 09:30 AM debido a una actualización programada. Por favor guarden su trabajo y completen las ventas activas antes de las 08:00 AM. Les notificaremos en cuanto el sistema esté disponible. Gracias por su comprensión.

7.2 Point of no return and stop criteria

If at 09:00 AM (30 minutes before window closure) the regression tests fail or core features (Login/POS) remain broken, execution stops immediately, rollback is applied, and the system is restored to v1.0.

Si a las 09:00 AM las pruebas de regresión fallan o las funciones principales (Login/POS) presentan errores, se detiene la ejecución, se aplica la reversión y se restaura la versión v1.0.

7.3 Success criteria

All regression tests pass, Cashiers are successfully restricted from product editing, the interface renders fully in English, and total test record counts match the baseline.

Todas las pruebas de regresión pasan, el rol Cajero no puede editar productos, la interfaz está completamente en inglés y el total de registros coincide con la línea base.

Approval / Aprobación

This plan is submitted for approval before execution. Execution may not begin until it is approved.

Este plan se somete a aprobación antes de la ejecución. No puede iniciarse la ejecución hasta ser aprobado.

Decision / Decisión	Approved ☐ Approved with conditions ☐ Returned for correction ☐
Conditions / Condiciones	

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